TEXORA · LECTURE SHEET

Cutting Calculation

Marker efficiency, lay planning and the cut-order plan: how an order quantity becomes metres of fabric, numbers of lays and plies, and spreading time — calculated end to end for a 12,000-piece shirt order.

AM-3207 · ADVANCED MACHINERIES AND RELEVANT CALCULATIONS · TOPIC 3 · SPECIALIZATION — APPAREL MANUFACTURING

1. The quantities a cutting room must know

Before a single ply is spread, the cutting department calculates four things: the **marker** (how the pattern pieces of several garments nest in one fabric width), the **lay plan** (how many plies are spread and cut together), the **fabric requirement** (total metres, net and gross), and the **cut-order plan** (how the order's size and colour quantities divide into lays). These calculations consume the pattern dimensions of Topic 2 and feed the fabric and cost calculations of Topics 5 and 6, so an error here is paid for in fabric.

2. Marker efficiency

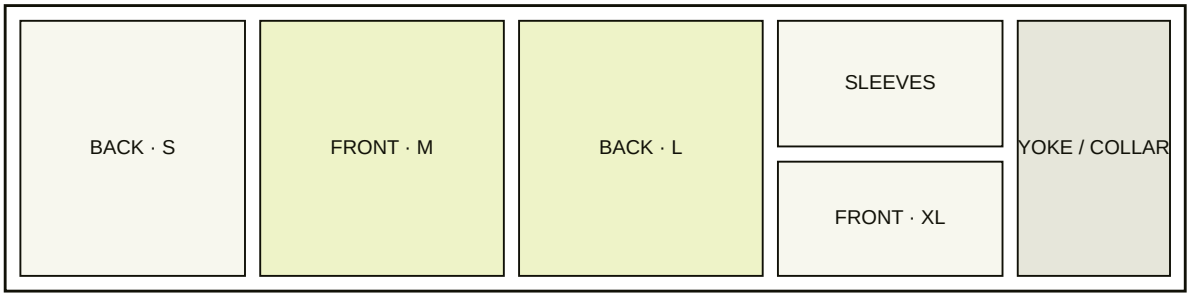
A **marker** is the arrangement of all pattern pieces for a group of garments within the usable fabric width. Its quality is measured by **marker efficiency**:

$$\text{Marker efficiency (\%)} = (\text{total pattern-piece area} \div \text{marker area}) \times 100$$

For the course order — 12,000 men's woven shirts in sizes S : M : L : XL = 1 : 2 : 2 : 1 — the marker carries one garment of each size combination per repeat: 6 garments (1 S, 2 M, 2 L, 1 XL) in a marker 5.40 m long on fabric of 1.50 m width, of which 1.47 m is usable between selvedges. The summed area of one garment's pattern pieces is 1.05 m².

WORKED EXAMPLE — MARKER EFFICIENCY

Pattern area per marker = 6 × 1.05 = 6.30 m². Marker area = 5.40 × 1.47 = 7.94 m².
Efficiency = 6.30 ÷ 7.94 × 100 = **79.4%**. The remaining 20.6% is inter-piece waste — the gaps between curved pieces — which is why marker making (CAD, Topic 1) is a profit centre: one extra point of efficiency on this order saves about 110 m of fabric.



Marker 5.40 m × 1.47 m usable width — 6 garments per marker · pattern area 6.30 m² of 7.94 m² = 79.4% efficiency

Figure 1 — Schematic marker plan (piece shapes simplified to blocks). Real markers nest the true curved outlines; the area arithmetic is identical.

3. The lay plan and fabric requirement

A **lay** is a stack of plies spread to the marker length and cut in one pass. Lay height is limited by the cutter and fabric (here, a maximum of 120 plies for this fabric on the CNC cutter). The lay plan follows directly:

WORKED EXAMPLE — CUT-ORDER PLAN FOR 12,000 SHIRTS

Garments per lay = 6 per marker × 120 plies = 720. Full lays = 16 × 720 = 11,520 garments. Remainder = 12,000 − 11,520 = 480 garments = 480 ÷ 6 = **80 plies** in a final part-lay. Total lays = **17**; total plies = 16 × 120 + 80 = **2,000 plies**.

Net fabric = 2,000 plies × 5.40 m = **10,800 m**, i.e. 10,800 ÷ 12,000 = **0.90 m per garment**.

Gross fabric with 3% cutting and end-loss allowance = 10,800 × 1.03 = **11,124 m**, or **0.927 m per garment** — the figure the costing of Topic 6 buys.

Size quantities check: S 2,000 · M 4,000 · L 4,000 · XL 2,000 = 12,000 ✓ (the 1:2:2:1 ratio inside every marker guarantees the ratio in every lay).

Spreading time at 0.5 min per ply by hand = 2,000 × 0.5 = 1,000 min ≈ **16.7 h** — compare the automatic spreader of Topic 1.

4. Losses the calculation must carry

LOSS	WHERE IT ARISES	HOW IT ENTERS THE CALCULATION
End loss	Each ply end must extend past the marker ends to be gripped and cut square	Added per ply, or inside the overall wastage percentage

Inter-piece waste	Gaps between pattern pieces	Already measured by marker efficiency — never added twice
Selvedge/width loss	Usable width is less than nominal width	Use usable width (1.47 m) in all area and efficiency work
Defect cut-outs / splice loss	Fabric faults are cut out and plies spliced during spreading	A fabric-quality-dependent allowance, part of the 3% here
Remnant (leftover) loss	Roll ends too short for a full ply	Minimised by roll planning against the lay plan

Glossary

TERM	MEANING
Marker	The arrangement of pattern pieces for a group of garments within the usable fabric width; its length sets fabric per garment.
Marker efficiency	Pattern-piece area as a percentage of total marker area.
Lay / ply	A ply is one layer of fabric in the stack; the stack cut together is the lay.
Cut-order plan	The division of an order's size/colour quantities into markers and lays.
End loss	Fabric consumed beyond the marker ends at each ply.

Key points and summary

- Marker efficiency = pattern area ÷ marker area; the course marker achieves 79.4% (6.30 of 7.94 m²).
- The lay plan turns order quantity into lays and plies: 12,000 shirts → 17 lays, 2,000 plies, 10,800 m net.
- Gross fabric adds cutting losses: 11,124 m at 3%, i.e. 0.927 m per garment — the costing figure.
- Keeping the size ratio inside the marker guarantees the ratio in every lay and simplifies the whole plan.

Practice

Answer in your own words; brief model answers follow.

1. A marker of 4 garments has pattern area 4.36 m², length 4.60 m and usable width 1.44 m. Calculate its efficiency.

2. An order for 9,600 garments uses the course marker (6 per marker) with lays of 100 plies. How many lays and total plies are needed, and how much net fabric?
3. Why must inter-piece waste not be added again as a separate loss after marker efficiency has been used?
4. The cutting wastage allowance rises from 3% to 5%. Recalculate gross fabric and gross consumption per garment for the course order.

Answer key

1. Marker area = 4.60 × 1.44 = 6.62 m². Efficiency = 4.36 ÷ 6.62 × 100 = 65.8%.
2. Per lay = 600 garments; 9,600 ÷ 600 = 16 lays exactly; plies = 16 × 100 = 1,600; net fabric = 1,600 × 5.40 = 8,640 m.
3. Marker efficiency already measures the inter-piece gaps: fabric bought = marker area, garments obtained = pattern area. Adding the gap a second time would double-count it.
4. Gross = 10,800 × 1.05 = 11,340 m; per garment = 11,340 ÷ 12,000 = 0.945 m.
